Can we detect CO?



Why molecular absorption has this shape and what the channel can reveal

60–1000 GHz | No receiver noise

15 October 2026

Can we detect it? The answer depends on what is known

Yes, in the exact model

CO produces a nonzero spectral change. A matched reference lets us recover its added absorption.

With an unrestricted unknown atmosphere, the split is not unique.

A practical detection limit still needs measured precision and a model of calibration and atmospheric errors.

The electric field can excite CO rotation

CO has a permanent electric dipole.
The wave couples to its rotational states.

$$E_J = h_P\, B\, J(J+1)$$

$$h_P\, f_0 = E_{J+1} - E_J$$

Allowed absorption: $J \rightarrow J + 1$

Transition	Center frequency
$0 \rightarrow 1$	115.271 GHz
$1 \rightarrow 2$	230.538 GHz
$2 \rightarrow 3$	345.796 GHz

CO ground vibrational state. Rigid rotor approximation; catalog frequencies include corrections.

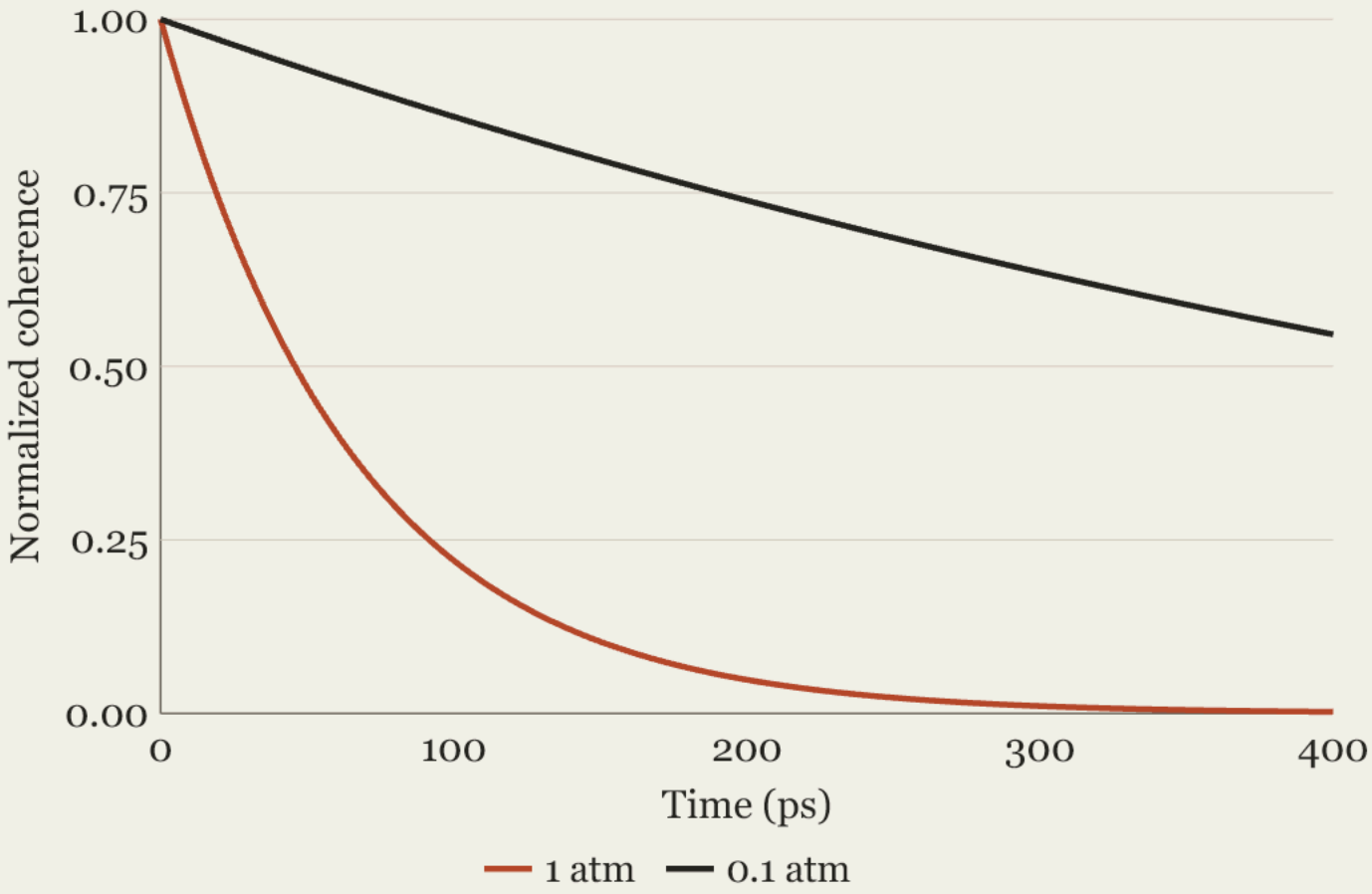
Collisions interrupt the molecular phase

Nearby molecules perturb the transition
and destroy phase coherence.

$$A(t) = \exp(-t / T_2)$$

Shorter coherence time
means a broader resonance.

At 1 atm: $T_2 \approx 66$ ps



Effective coherence times inferred from the first CO line's air broadened width at 296 K.

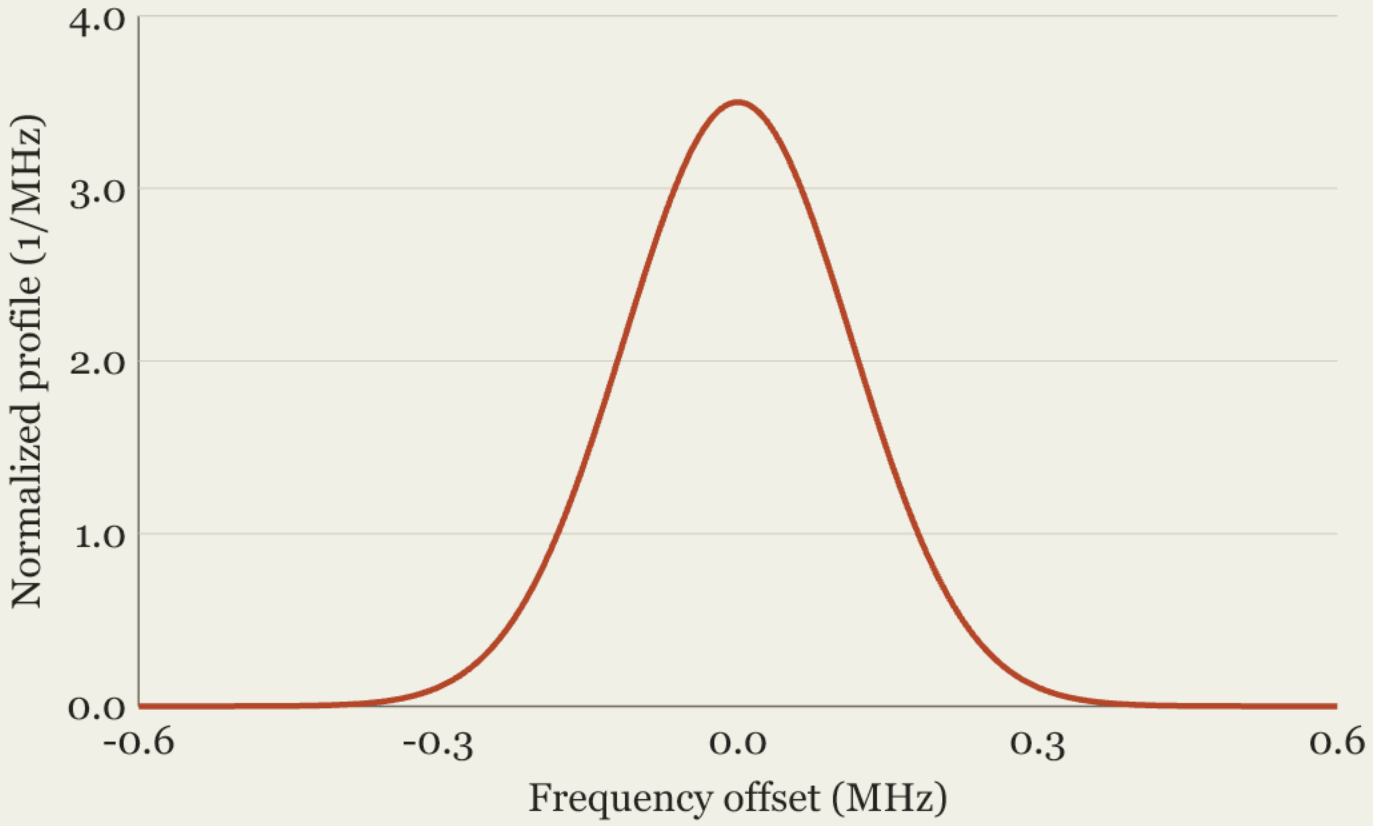
Thermal motion gives different Doppler shifts

$$\Delta f \approx f_o \, v / c$$

Molecules have different velocities along the beam.

Thermal motion produces a Gaussian frequency profile.

Doppler full width: 0.268 MHz



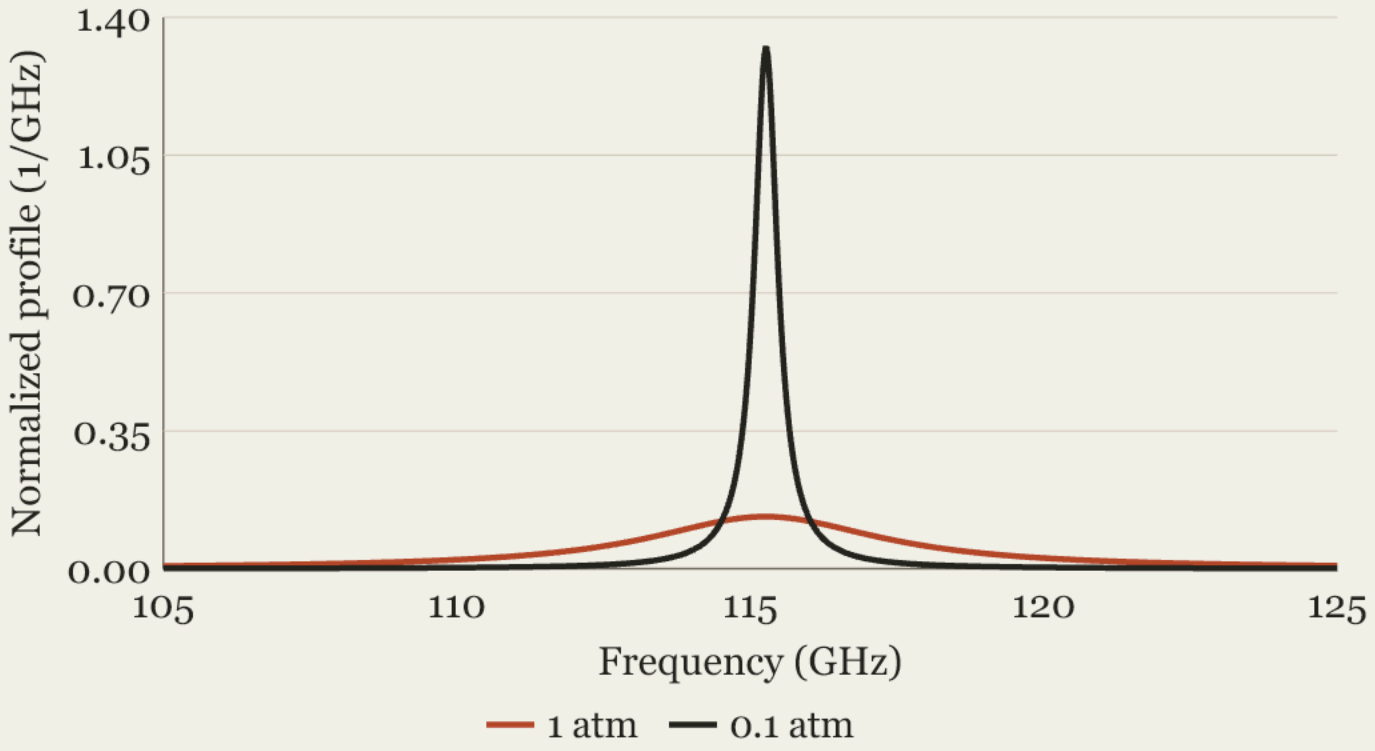
First CO line at 115.271 GHz and 296 K. The horizontal axis is in MHz.

At atmospheric pressure, collisions set most of the width

Voigt profile = Lorentz * Gaussian

Pressure broadening: 4.815 GHz
Doppler broadening: 0.000268 GHz

For this line at 296 K and 1 atm, the pressure width is about 18,000 times larger.



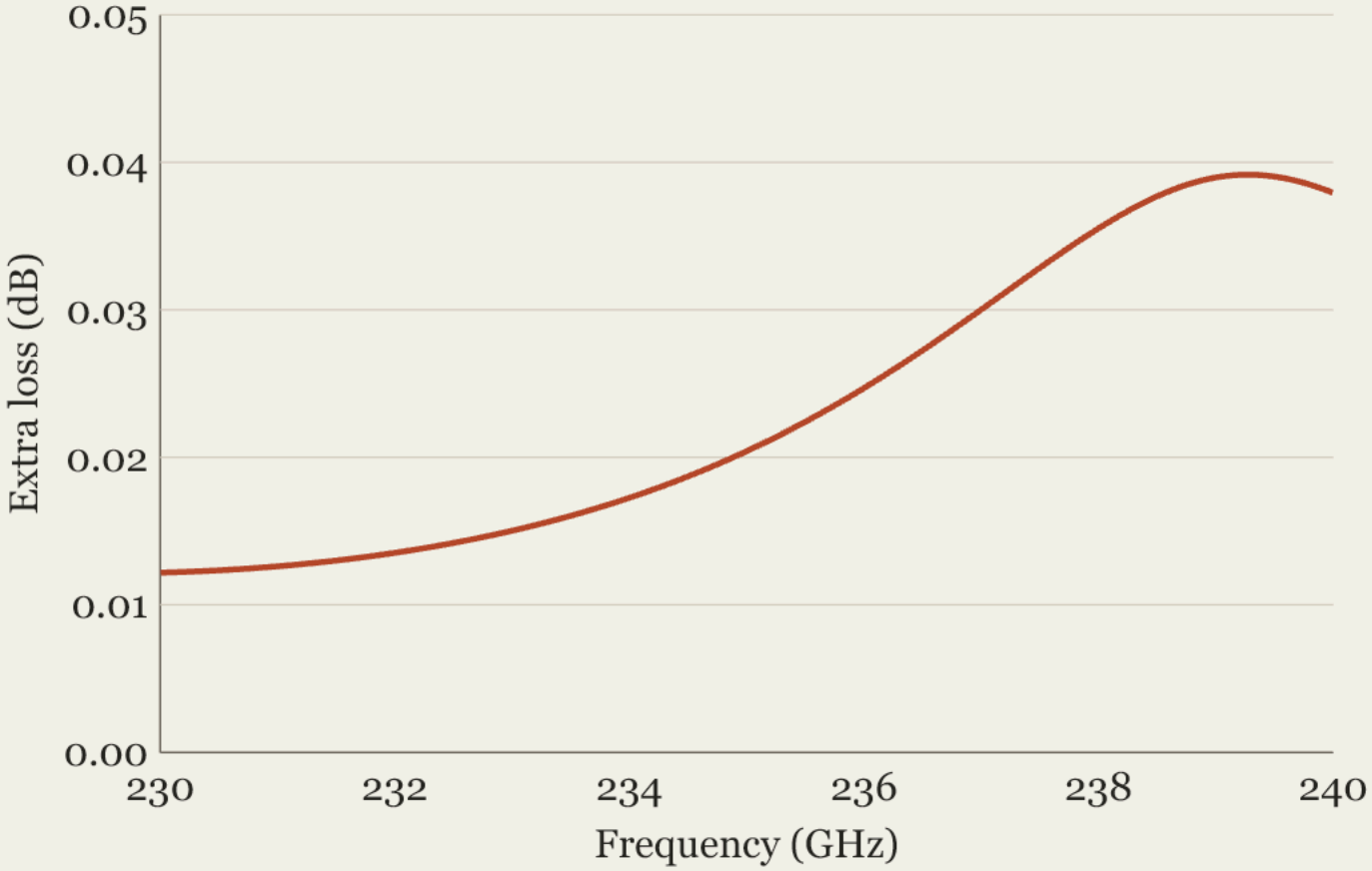
Same integrated line area. This comparison isolates broadening, not equal ppm samples.

Why the curve in your image is smooth

The original calculation adds CH_3CN line contributions along an atmospheric path.

Broad wings overlap. Different pressures along the path also change their widths.

A smooth shape alone does not prove a separate continuum.



Reconstructed historical CH_3CN model curve. It is not a CO spectrum or a measured result.

Three physical steps connect a molecule to the channel

1 Sum the spectral lines

$$\sigma(f) = \Sigma_j \, S_j \, V_j(f - f_j)$$

Line strength × normalized profile

2 Accumulate along the path

$$\tau_c(f,l) = \int n_c(s) \, \sigma(f;s) \, ds$$

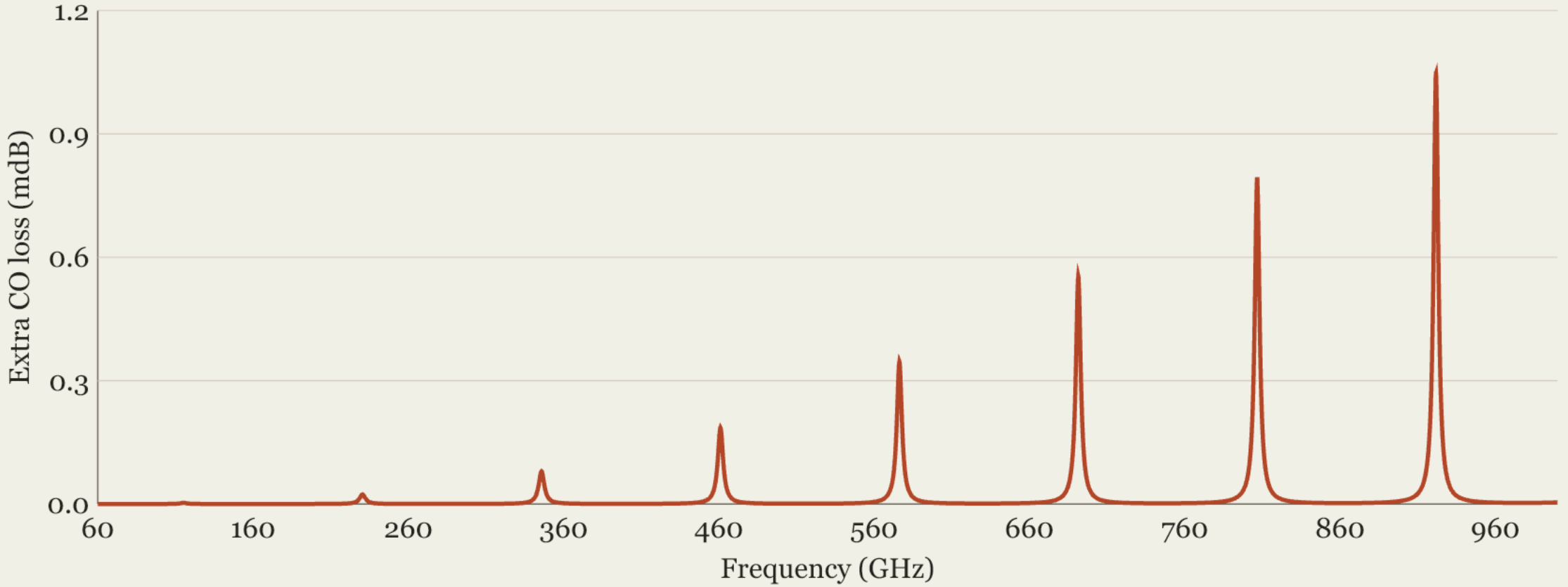
Integrate density × cross section from 0 to l

3 Attenuate the field

$$|a_c| = \exp(-\tau_c / 2)$$

Power transmission is exp(−τc).

CO provides a sequence of spectral features



Eight main ground state lines in this band. Heights depend on populations, strengths and widths.

Predicted CO loss: 1 ppm over 20 m, 296 K and 1 atm. Main isotopologue. Exact model.

Observe the chemical loss through a channel ratio

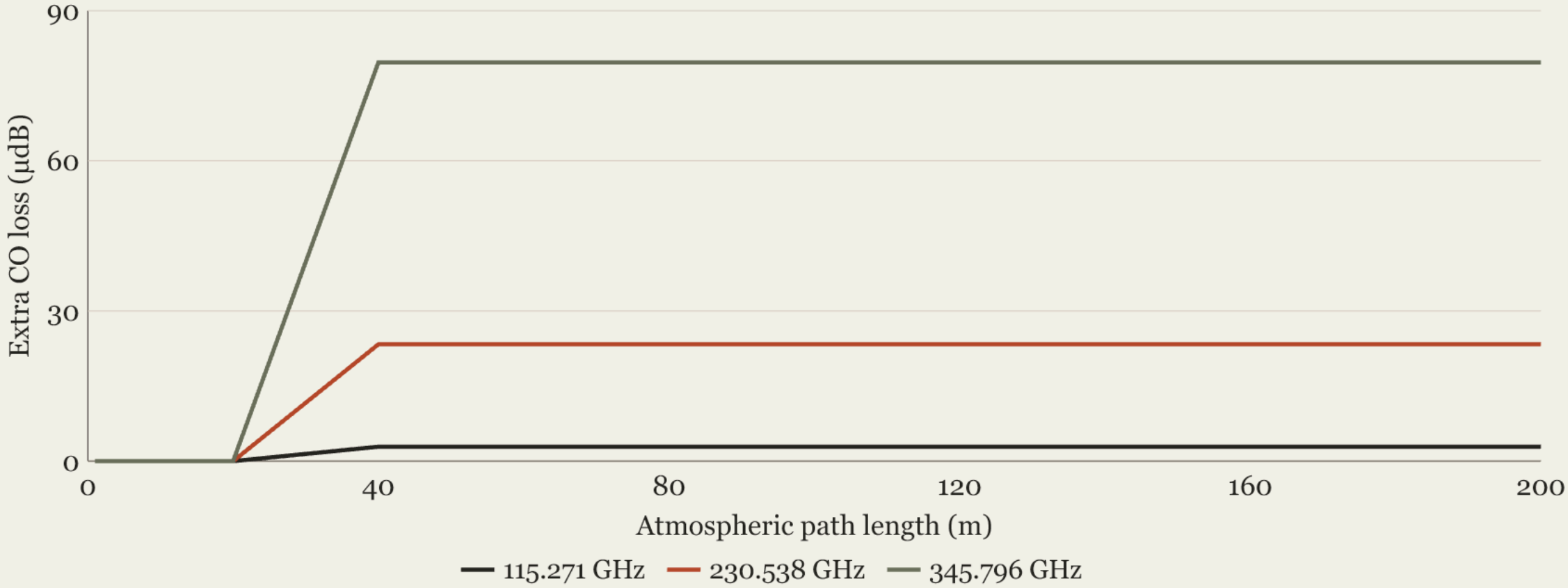
Measure $h(f,l)$. Obtain a matched reference $h_0(f,l)$.

$$|a_c| = |h| \,/\, |h_0|$$

$$\tau_c = -2 \ln(|h| \,/\, |h_0|)$$

The reference must preserve geometry, background atmosphere and instrument response. It excludes the added CO.

Path length changes the gas column that the wave crosses



CO between 20 and 40 m: loss grows inside the cloud, then stops growing.

h(f,l): frequency × atmospheric path length. Nested paths through one stationary medium.

The predicted signal is small, even when it is identifiable

Relative drop in channel amplitude, before receiver errors

Frequency	Amplitude change	Atmospheric loss
115.271 GHz	0.33 ppm	0.15 dB
345.796 GHz	9.17 ppm	1.69 dB
576.268 GHz	39.80 ppm	99.53 dB
921.800 GHz	120.65 ppm	38.46 dB

These ppm describe amplitude change, not gas concentration.
A larger chemical peak can come with much greater background loss.

1 ppm added CO over 20 m. Atmospheric path 200 m, 296 K, 1 atm, water partial pressure 1000 Pa.

A small atmospheric mismatch can dominate the CO loss

1 ppm CO across 20 m

0.0000797 dB

Water partial pressure changes from 1000 to 1010 Pa across 200 m

0.0187 dB

About 235 × larger. Their spectral shapes may still allow separation.

Chosen sensitivity example at 345.796 GHz. Fixed 296 K and 1 atm. No receiver noise.

CO must be distinguishable from atmospheric changes

$$r(f,l) = \tau_m(f,l) + \tau_c(f,l)$$

Unknown, unrestricted atmosphere

Many splits of the same total loss produce the same channel.

Constrained atmosphere

Fit the frequency pattern jointly.
CO must add a response that the allowed background cannot reproduce.

We need temperature, pressure, humidity, reference CO and instrument response.

What we can conclude now

CO is observable in the ideal matched case.

Its smooth spectrum follows from molecular transitions, finite coherence and thermal motion.

To claim practical detection, establish the reference accuracy and test whether the CO response survives the allowed errors.

Appendix: the law behind the smooth line

Exponentially decaying molecular coherence

$$C(t) = \exp(-t / T_2) \exp(i\, 2\pi\, f_0\, t)$$

Its frequency response has a Lorentz absorption profile

$$L(\Delta f) = \gamma / [\pi(\Delta f^2 + \gamma^2)]$$

$$\gamma = 1 / (2\pi\, T_2)$$

Finite coherence gives finite width. Doppler shifts convolve this line with a Gaussian.
An infinitely sharp ideal transition would be a delta peak, not a rectangular band.

Sources and numerical evidence

NIST molecular data	CO dipole and rotational selection rules
MIT spectroscopy	Coherence, dephasing and line shape
HITRAN and NIST MATS	Line parameters, Doppler and collision profiles
JPL CO catalog	Independent line positions
ITU P.676, Annex 1	Atmospheric attenuation through 1000 GHz
Project calculation	Assumptions, spectra and channel matrices

Scientific qualifications and additional source links are included in the speaker notes.